Brain MRI Tissue Segmentation with Traditional Computer Vision: 2D and 3D Approaches

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Abstract

We study tissue segmentation of T1-weighted brain MRI using only traditional computer-vision techniques, partitioning each slice into six anatomical regions: air, skin/scalp, skull, cerebrospinal fluid (CSF), gray matter, and white matter. Three 2D pipelines are developed, all sharing a common structure — brain extraction by active contours, followed by independent segmentation of the outer (non-brain) and inner (brain) regions through Otsu thresholding and either K-Means clustering or multi-Otsu thresholding. The pipelines are compared against ground-truth labels using a weighted F1 (Dice) score complemented by the Jaccard index, and the strongest approaches are extended to operate on the full 3D volume. Replacing Morphological Chan–Vese (MCV) with Morphological Geodesic Active Contours (MGAC) for brain extraction improves the CSF F1 score by 0.23 and raises the weighted average from 0.91 to 0.92; K-Means gives the best balance of accuracy and reliability while multi-Otsu is marginally faster; and the 3D extension yields a further small gain (weighted F1 0.93) by exploiting inter-slice continuity, at roughly 2.3× the computational cost.

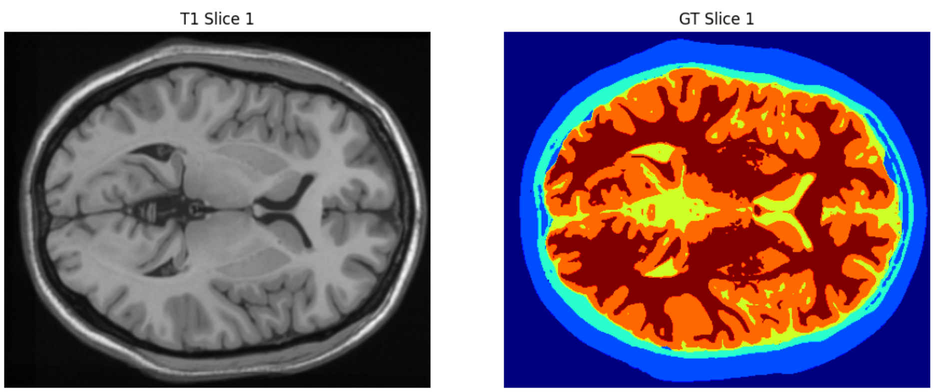
1. Introduction

Accurate delineation of brain tissue in magnetic resonance imaging (MRI) is a core step in many clinical and research pipelines. While modern approaches rely heavily on deep learning, classical computer-vision methods remain attractive when annotated data and computational resources are scarce: they are interpretable, require no training, and run on a CPU. This report investigates how far such traditional techniques can be pushed on a tissue-segmentation task.

The data consist of ten consecutive axial cross-sections of a publicly available T1-weighted MRI of a single human subject, each of size 362 × 434 pixels, acquired at 1.00 mm intervals (10.00 mm total span). Pre-segmented ground-truth labels accompany the images and define six tissue classes, summarised in Table 1. Figure 1 shows a representative input slice alongside its ground truth.

Table 1: The six tissue classes and their integer labels.

Label	Tissue
0	Air
1	Skin / scalp
2	Skull
3	CSF (cerebrospinal fluid)
4	Gray matter
5	White matter



(a) T1-weighted input

(b) Ground-truth labels

Figure 1: A representative axial slice: the raw T1-weighted image (left) and the ground-truth six-class segmentation (right).

We pursue three goals. First, we develop several 2D segmentation algorithms that consistently partition each slice into the six regions (Section 2). Second, we evaluate and compare them with a well-justified accuracy metric and identify the most robust algorithm (Sections 3–4). Third, we extend the best-performing methods to process the entire volume at once and contrast the 3D and 2D behaviour (Section 5).

2. 2D Segmentation Methods

All three pipelines follow the same four-step structure: (1) separate the outer regions (labels 0–2) from the inner brain (labels 3–5) by extracting a brain mask; (2) segment the outer regions 0–2; (3) segment the inner regions 3–5; and (4) combine the partial results into a complete label map. The methods differ in how the brain is extracted in step (1) and in the technique used for the inner regions in step (3).

2.1 Method 1: Morphological Chan–Vese + Otsu + K-Means

For brain extraction we use the Morphological Chan–Vese (MCV) technique [2, 4]. This variational method evolves a contour that captures homogeneous regions by minimising an energy functional, without relying explicitly on gradient information. After the initial binary brain mask is obtained, a *closing* operation connects boundaries across thin indentations (forming enclosed holes), and an *area closing* operation then refines the region by filling those gaps.

For the outer regions, Otsu thresholding [1] first separates labels 0 and 2 from labels 1, 3, 4, 5. To isolate the skin/scalp (label 1), Otsu thresholding is applied to the T1 image with the brain cropped out. To obtain the skull (label 2), all non-zero intensities in the T1 image are raised to the maximum to form a binary mask covering labels 1 and 2, from which the label-1 mask is subtracted. For the inner brain (step 3), K-Means clustering [5] partitions the region into labels 3, 4, and 5. Finally, all regions are merged (step 4).

Discussion. MCV manages to crop the brain region, but unevenly. The closing operation helps only marginally — mostly with small, thin indentations — whereas area closing proves effective at filling the larger gaps. The method is robust, but a more accurate brain-extraction technique in step (1) is required for further progress. Only two `area_threshold` values were chosen heuristically for area closing, set as soon as the largest gap in the brain region was closed.

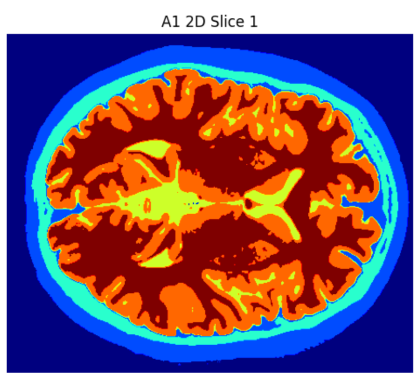


Figure 2: Method 1 (MCV + Otsu + K-Means), slice 1.

2.2 Method 2: Morphological Geodesic Active Contours + Otsu + K-Means

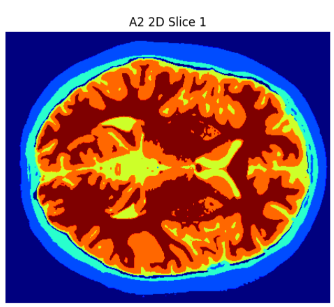
Here the MCV brain extraction of step (1) is replaced by its variant, Morphological Geodesic Active Contours (MGAC) [3, 4]. Whereas MCV relies on regional homogeneity — present in the brain region, but not to the desired extent — MGAC uses an edge indicator derived from image gradients to guide the evolving contour. In our case the target boundary (the skull, label 2) has intensities near zero, so on the raw T1 images the contour naturally evolves to the desired edge; a balloon force further drives the contour outward until it reaches the skull. Steps (2)–(4) are unchanged from Method 1.

Discussion. MGAC successfully crops the brain region with visibly better results than MCV. With the brain-extraction step improved, we can build on it in the subsequent steps and attempt to make the pipeline even more robust.

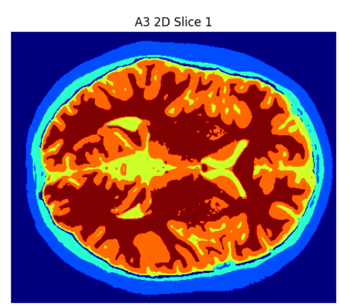
2.3 Method 3: MGAC + Otsu + Multi-Otsu

As an experiment, in step (3) multi-Otsu thresholding replaces K-Means. Multi-Otsu may be more robust because it computes thresholds from the global histogram, reducing sensitivity to initialisation and noise. However, multi-Otsu fails when histogram peaks are indistinct due to noise or overlap (not the case here), whereas K-Means — clustering pixels by similarity rather than relying on the histogram — can handle more complex distributions. Steps (1), (2), and (4) are identical to Method 2.

Discussion. Visually, multi-Otsu performs as well as K-Means; further quantitative evaluation is required. Although the segmentation could likely be improved further (small gaps remain between regions 2 and 3), doing so — e.g. by experimenting with region growing or flood fill — would require more parameter tuning (seeds, thresholds) and make the algorithm more sensitive to intensity variation, and therefore less robust.



(a) Method 2, slice 1



(b) Method 3, slice 1

Figure 3: Single-slice 2D results for Method 2 (MGAC + K-Means) and Method 3 (MGAC + multi-Otsu).

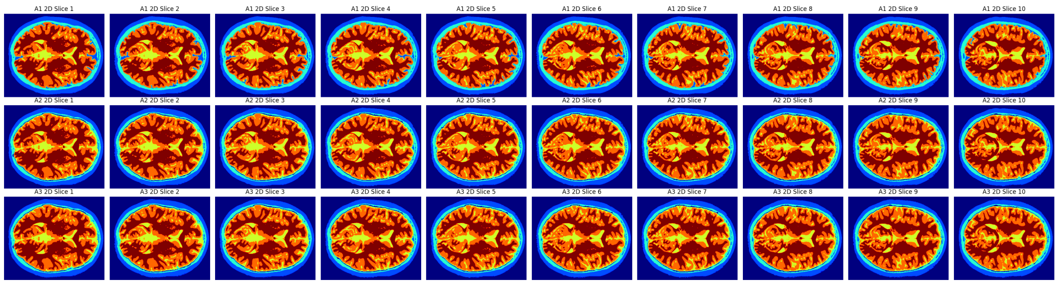


Figure 4: All ten slices for the three 2D methods. Top to bottom: Method 1 (MCV + K-Means), Method 2 (MGAC + K-Means), Method 3 (MGAC + multi-Otsu).

3. Evaluation Metric

We evaluate the segmentations against the ground-truth labels primarily with the *weighted F1 (Dice) score*, and report the Jaccard index alongside it. Unlike the Jaccard index — which measures overlap as the ratio of intersection to union and is very sensitive to minor boundary discrepancies — the F1 score emphasises the balance between precision and recall. Its harmonic-mean formulation accounts for false positives and false negatives without disproportionately penalising small regions. Because the tissue regions here are highly imbalanced, the weighted F1 score is the more appropriate primary metric, with the Jaccard index reported for completeness.

4. 2D Results and Discussion

Overall and per-region scores for the three methods are reported in Tables 2 and 3.

Table 2: Overall 2D results: weighted F1, weighted Jaccard index, and mean execution time over five runs.

Approach	F1	Jaccard	Time (s)
A1 (MCV + K-Means)	0.91	0.850	56.86
A2 (MGAC + K-Means)	0.92	0.862	62.27
A3 (MGAC + Multi-Otsu)	0.92	0.858	60.66

Table 3: Per-region 2D scores (F1 and Jaccard J). A1: MCV + K-Means; A2: MGAC + K-Means; A3: MGAC + Multi-Otsu.

Region	Tissue	A1	A2	A3
0	Air	F1=0.97, J=0.937	F1=0.95, J=0.900	F1=0.95, J=0.899
1	Skin/scalp	F1=0.82, J=0.700	F1=0.87, J=0.774	F1=0.87, J=0.774
2	Skull	F1=0.84, J=0.732	F1=0.74, J=0.593	F1=0.74, J=0.593
3	CSF	F1=0.64, J=0.471	F1=0.87, J=0.769	F1=0.87, J=0.775
4	Gray matter	F1=0.94, J=0.889	F1=0.94, J=0.895	F1=0.94, J=0.888
5	White matter	F1=0.98, J=0.957	F1=0.97, J=0.943	F1=0.97, J=0.933

Method 1 does a poor job segmenting the CSF (region 3) — a consequence of using MCV for brain extraction in step (1) — while performing well on the other regions (0, 1, 2, 4, 5).

Method 2 shows a significant 0.23 improvement on region 3, attributable to MGAC's superior separation of the outer (0–2) and inner (3–5) tissues. The remaining regions are segmented well.

Method 3 shows no improvement over Method 2: K-Means and multi-Otsu yield essentially identical scores, because the inner brain has a homogeneous intensity distribution — a narrow, distinct range of values relative to adjacent tissues — where both techniques excel.

Choosing the best method. The choice of MGAC over MCV produces a large improvement on region 3, whose border has a distinct gradient that MGAC exploits. Methods 2 and 3 are very close: they are identical on weighted F1 (0.92), while on the Jaccard index K-Means (Method 2) is marginally better (0.862 vs 0.858) and multi-Otsu (Method 3) is slightly faster (60.66 s vs 62.27 s, Table 2). Method 2 therefore offers the best balance of accuracy and reliability, whereas Method 3 is preferable when speed is the priority. Being the two strongest 2D pipelines, both are carried forward to 3D.

5. 3D Segmentation

The two best-performing 2D approaches (Method 2 and Method 3, both built on MGAC) were extended to process the entire MRI volume simultaneously rather than slice by slice. Overall and per-region scores are given in Tables 4 and 5.

Table 4: Overall 3D results: weighted F1, weighted Jaccard index, and mean execution time over five runs.

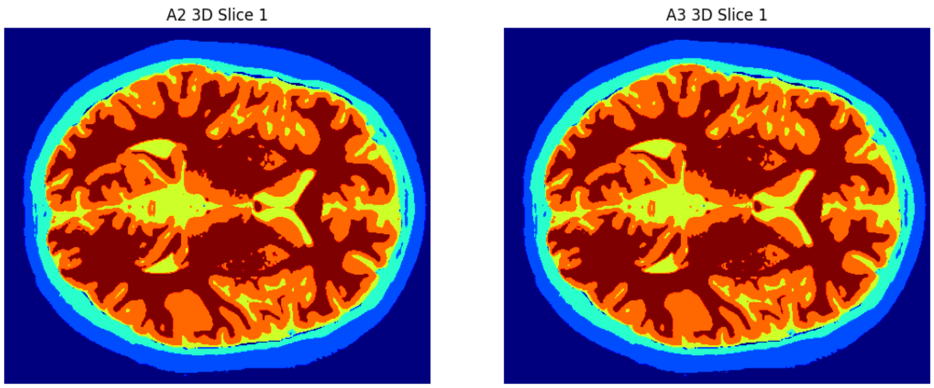
Approach	F1	Jaccard	Time (s)
A2-3D (MGAC + K-Means)	0.93	0.867	144.69
A3-3D (MGAC + Multi-Otsu)	0.93	0.865	141.11

Table 5: Per-region 3D scores (F1 and Jaccard J). A2-3D: MGAC + K-Means; A3-3D: MGAC + Multi-Otsu.

Region	Tissue	A2-3D	A3-3D
0	Air	F1=0.95, J=0.906	F1=0.95, J=0.905
1	Skin/scalp	F1=0.87, J=0.771	F1=0.87, J=0.771
2	Skull	F1=0.77, J=0.627	F1=0.77, J=0.627
3	CSF	F1=0.87, J=0.774	F1=0.88, J=0.778
4	Gray matter	F1=0.95, J=0.898	F1=0.94, J=0.895
5	White matter	F1=0.97, J=0.947	F1=0.97, J=0.943

Both 3D approaches perform slightly better than their 2D counterparts — a 0.01 increase in weighted-average F1 (to 0.93), driven mainly by a 0.03 improvement on region 2 (skull), which is practically the hardest area to segment because it shares the same intensity range as region 0. No region's score declines. We attribute this to the fact that the 3D approach leverages inter-slice continuity: even where region 2's intensity overlaps with region 0, the additional spatial context helps refine the boundary delineation. This is the principal advantage of the 3D algorithms.

However, the 3D pipelines are markedly slower: averaged over five runs they take 144.69 s (A2-3D) and 141.11 s (A3-3D), roughly 2.3× their 2D counterparts — the main disadvantage of operating on the full volume. Adapting and tuning algorithms for 3D can also be more complex, particularly when inter-slice alignment is imperfect (precisely where the deliberate focus on robustness in 2D begins to matter). Finally, although spatial context reduces slice-to-slice variability, it is a double-edged sword: segmentation errors can propagate across the volume.



(a) Method 2 in 3D, slice 1 **(b)** Method 3 in 3D, slice 1

Figure 5: Single-slice results from the 3D pipelines.

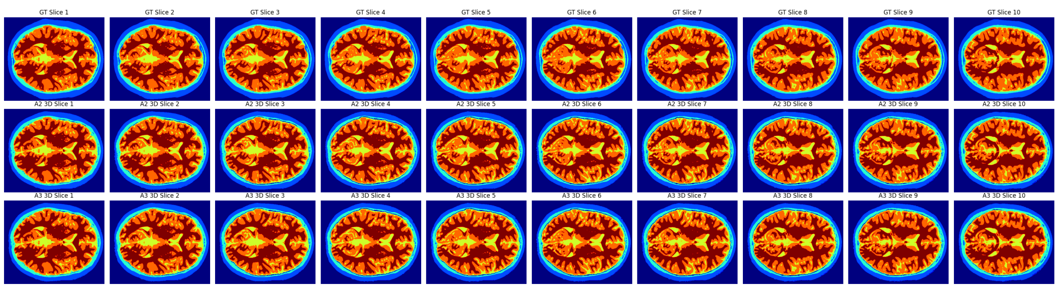


Figure 6: All ten slices in 3D. Top to bottom: ground truth, Method 2 in 3D, Method 3 in 3D.

6. Conclusion

Using only traditional computer-vision techniques, we built three 2D pipelines for six-class brain-tissue segmentation and extended the best-performing ones to 3D. Replacing MCV with MGAC for brain extraction was the single most impactful change, substantially improving CSF (region 3) segmentation thanks to the distinct gradient at the brain boundary. Our main findings are:

1. **Best 2D approach.** Approach 2 (MGAC + K-Means) provides the best balance of accuracy and reliability.

2. **3D advantage.** Processing the full volume significantly improves segmentation quality, especially for challenging regions, by leveraging inter-slice continuity (weighted F1 0.93 vs 0.92).

3. **Method selection.** For accuracy, prefer Approach 2 (in 2D or 3D); for speed, use multi-Otsu thresholding (Approach 3); for robustness, apply 3D processing

when inter-slice alignment is good.

4. **Practical considerations.** 3D approaches are more computationally expensive (roughly 2.3×) but deliver better results, which is valuable for clinical applications.

The choice between 2D and 3D is therefore a trade-off between performance gains and computational efficiency.

Reproducibility. The implementation of all methods is available in the accompanying project repository.

References

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